

10	Determination of resistivity by four probe method.	01	LO5
11	Any other experiment based on syllabus may be included, which would help the learner to understand concept. Virtual lab may be developed and used for performing the experiments , after defining a suitable LO	01	LO6

Term Work Marks: 25 Marks (Total marks) = 10 Marks (Experiment) + 10 Marks Project + 5 Marks (Attendance)

Project work will be extended to semester-2 as well. In semester 1, a group of four students will be formed; a domain may be provided by faculty, the group will frame a problem statement in consultation with faculty. A PPT presentation with problem statement, preliminary literature survey, execution plan and a probable outcome is to be considered for awarding marks. Proper rubrics must be framed by faculty member

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
BSC103	Applied Chemistry	2	--	-	2	--	-	2

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment Test (IAT)			End Sem Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	IAT-I + IAT-II (Total)					
BSC103	Applied Chemistry	15	15	30	45	02	--	--	75

Rationale:

Chemical science has contributed in many ways to most of the Engineering branches where “Environmental Chemistry” is the modern approach to learn impact of Technology on habitat and can be common to all Core Groups, “Engineering Materials” can be prerequisites to many subjects of all core groups and Impact of corrosion on metals as engineering materials is the important area of concern. “Conventional and Non Conventional Energy Study” is the matter of general approach to all Core groups as Energy issue is the most recent concern even for designing computational engines (Include hardware & software energy efficient).

Course Objectives:

- 1) To study Coal as a conventional source of energy.
- 2) To study the effect of corrosion by different mechanisms on metals and methods of corrosion control.

- 3) To recognise importance of alloys and can apply the phase rule on it to study the effect of temperature and composition.
- 4) To introduce important properties of polymers as Engineering material.
- 5) To recognise the composition, properties and functions of various composite materials.
- 6) To study importance of Green Chemistry by comparative study of conventional and Green routes of syntheses, solvents and fuels.

Course Outcomes: Student will be able to -

- 1) Determine the quality of coal and quantify the oxygen required for combustion of coal.
- 2) Apply different methods to minimize corrosion in industries.
- 3) Interpret various phase transformations of alloy using thermodynamics.
- 4) Use the polymers for specific engineering applications on the basis of the properties.
- 5) Identify different types of composite materials for engineering applications.
- 6) Apply the principles of Green chemistry and study environmental impact for sustainable development

Prerequisite:

- 1) Knowledge about basic difference in Conventional and non-conventional energy sources.
- 2) Knowledge about concepts of Electrochemistry.
- 3) Knowledge of basic properties of metals and nonmetals.
- 4) Knowledge of 12 principles of Green Chemistry

DETAILED SYLLABUS:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
I	Fuels and Combustion	A) cFuel: - Definition, Characteristics of good fuel. B) Calorific value (Definition, Types, Determination, Dulong's formula, Numerical) C) Coal: - Analysis of coal – Proximate analysis, Ultimate analysis, Numerical) D) Combustion of coal – Numerical	04	CO1
II	Corrosion	A) Introduction: - Definition, Types of Corrosion – i) Dry or Atmospheric Corrosion, ii) Wet or Electrochemical corrosion (In Acidic medium, In Neutral medium) B) Factors affecting rate of corrosion:- i) Position of metal in galvanic series, ii) Purity of Metal, iii) Nature of Corrosion product, iv) Temperature, v) pH of medium, vi) concentration of medium, vii) moisture, viii) Relative Cathodic and Anodic area, ix) overvoltage C) Methods to control corrosion: - i) Selection of metal, ii) Proper Designing, iii) Cathodic protection, iv) Use of Corrosion Inhibitors, v) Metallic Coating D) Corrosion in Electronic devices	04	CO2
III	Alloys	A) Purpose of making alloys. B) i) Gibbs Phase rule – Statement, Terms involved	04	CO3

		with examples. ii) Reduced phase rule, Two-component system (Pb-Ag) & Numerical. iii) Merits and Limitations of Phase rule.		
IV	Introduction to Polymers	A) Macro-molecular science, basic concept of polymers, Chemical bonding in polymers, Classification of Polymers. B) Properties of Polymers:- i) Molecular weight - Number average molecular weight, Weight average molecular weight, Numerical, ii) Crystallinity - Crystalline and amorphous polymers – Glass transition temperature, iii) Mechanical Properties: Hardness, tensile strength, creep, fatigue, impact resistance (introduction), iv) Electrical properties: dielectric strength, insulation resistance, surface resistivity (Introduction), v) Optical properties: refractive index, transmittance, photoelectric property, colour	05	CO4
V	Introduction to Composites	A) Definition, Characteristics of Composites, B) Constituents of Composites – Matrix Phase and Dispersed Phase (Definition and Functions) C) Classification of Composites	04	CO5
VI	Green Chemistry for sustainable development	A) Comparative study of synthesis of following industrially important molecules by conventional and green route:- i) Indigo dye, ii) Adipic acid, iii) Carbaryl B) Green Solvents: - characteristics and applications of Supercritical solvents and ionic liquids C) Green Fuels:- Synthesis and Advantages of i) Biodiesel, ii) Ethanol	05	CO6

Recommended Books:

1. Engineering Chemistry, Jain and Jain, Dhanpat Rai Publication
2. A textbook of Engineering Chemistry, S. S. Dara, S. Chand and Company
3. Polymer science: Vasant Gowarikar, Wiley Estern Ltd, new Delhi
4. Green Chemistry: V. K. Ahluwalia

Online References:

Sr. No.	Website Name
1.	https://archive.nptel.ac.in/courses/103/106/105106205/
2.	https://courses.nptel.ac.in/noc20_ch41/preview

Assessment:

Internal Assessment Test (IAT) for 15 marks each:

- IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

End Semester Theory Examination:

Question paper format

- Question Paper will comprise a total of **five questions each carrying 15 marks** Q.1 will be **compulsory** and should **cover the maximum contents of the syllabus**
- Remaining questions** will be **mixed in nature** (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of **three questions** need to be answered.

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
BSL102	Applied Chemistry Lab	--	1	-	--	0.5	-	0.5

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment Test (IAT)			End Sem Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	IAT-I + IAT-II (Total)					
BSL102	Applied Chemistry Lab	--	--	--	--	--	25	--	25

Lab Objectives:

- To apply knowledge acquired during the theory class in carrying out the experiments for qualitative and quantitative determination.
- To analyse experimental results and write laboratory report.

Lab Outcomes: After completion of experiment, the learners will be able to:

- Understand the significance of proximate analysis of coal and determine quality of coal sample.
- Learn various quantitative analytical techniques to determine % of elements from alloy samples.
- Synthesize biodiesel at laboratory level and calculate % atom economy from Green chemistry point of view.

4. Learn the effect of various factors on the rate of corrosion.
5. Synthesize bioplastic at laboratory level using from Green chemistry.
6. Quantitative determination of N₂ / Flue gas.

Prerequisite:

1. Knowledge of basic safety practices in the Chemistry Laboratory
2. Knowledge of Proximate analysis of coal
3. Knowledge of volumetric analysis

List of Experiments.

Sr No	List of Experiments	Hrs	LO Mapping
01	Determination of moisture content of coal	01	LO1
02	Determination of ash content of coal	01	LO1
03	Determination of Zn in Brass	01	LO2
04	Synthesis of Biodiesel from vegetable oil	02	LO3
05	Determination of Cu in Brass	01	LO2
06	Flue gas analysis by Orsats Apparatus	02	LO6
07	Synthesis of biodegradable plastics	02	LO5
08	Determination of nitrogen by Kjeldahl's method	02	LO6
09	To compare rate of corrosion of various metals in acidic medium	01	LO4

Sr No	List of Assignments / Tutorials	LO Mapping
01	Numerical based on calorific value determination, proximate and ultimate analysis of coal	LO1
02	Phase Diagram on Electrochemical corrosion in different medium	LO4
03	Diagrams and numerical based on two component system	LO2
04	Numerical based on average molecular weight of polymers	LO5
05	Synthesis of at least two Industrially important molecules	LO3, LO5

Assessment :

Term Work: Term Work shall consist of at least 5 to 6 practicals based on the above list. Also, Term work Journal must include at least 4 assignments.

Term Work Marks: 25 Marks (Total marks) = 15 Marks (Experiment) + 5 Marks (Assignments) + 5 Marks (Attendance)

Course Code	Course Name	Teaching Scheme (Contact Hours)			Credits Assigned			
		Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
ESC101	Engineering Mechanics	02	-	-	02	-	-	02

Course Code	Course Name	Theory					Term work	Pract / Oral	Total
		Internal Assessment Test (IAT)			End Sem Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	IAT-I + IAT-II (Total)					
ESC101	Engineering Mechanics	20	20	40	60	02	--	--	100

Rationale:

Engineering mechanics is a branch of science that deals with the behavior of solid bodies when subjected to external forces or loads and the effects of these forces on the bodies. It is a fundamental discipline within engineering and provides the basis for understanding and analyzing various types of structures and mechanisms.

Course Objectives:

1. To acquaint with basic principles of centroid and its application
2. To familiarize with the concepts of force, moment, Resultant and Equilibrium of system of coplanar force.
3. To acquaint with the basic concept of friction and its application in real-life problems.
4. To understand the parameters required to quantify the Kinematics of Particle and Rigid body.
5. To understand the parameters required to quantify the Kinetics of rigid body.
6. To acquaint with the basics of Robot kinematics

Course Outcomes:

1. Determine the equivalent force-couple system for a given system of forces.(L3)
2. Demonstrate the understanding of Centroid and its significance and locate the same. (L3)
3. Illustrate the concept of force, moment and apply the same along with the concept of equilibrium in two- and three-dimensional systems with the help of FBD. (L3)
4. Calculate position, velocity and acceleration etc. of particle/rigid body using principles of kinematics (L3)
5. Analyze particles in motion using force and acceleration, work-energy and impulse-momentum principles (L4)